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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/284,747	09/28/2023	Shotaro TOYA	Q292131	2161

23373 7590 09/10/2026

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EXAMINER	
DICUS, TAMRA	

ART UNIT	PAPER NUMBER
1787	

NOTIFICATION DATE	DELIVERY MODE
09/10/2026	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 18/284,747	Applicant(s) TOYA et al.	
	Examiner TAMRA L DICUS	Art Unit 1787	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 08/17/2026.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-3,5-6,8-9,12-13,15-16,18-21 and 23 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-3,5-6,8-9,12-13,15-16,18-21 and 23 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☐ The drawing(s) filed on _____ is/are: a)☐ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☑ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☑ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☑ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☐ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

A request for continued examination under 37 CFR 1.114, including the fee set forth in 37 CFR 1.17(e), was filed in this application after final rejection. Since this application is eligible for continued examination under 37 CFR 1.114, and the fee set forth in 37 CFR 1.17(e) has been timely paid, the finality of the previous Office action has been withdrawn pursuant to 37 CFR 1.114. Applicant's submission filed on 08/17/2026 has been entered.

Applicants' arguments have been fully considered. Rejections and/or objections not reiterated from previous office actions are hereby withdrawn due to Applicant's amendments and/or arguments. The following rejections and/or objections are either reiterated or newly applied.

Claim Rejections - 35 USC § 103

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

Claims 1-4, 7-13, and 19-20 are rejected under 35 U.S.C. 103 as being unpatentable over US 20110217541 to Shimano et al. in view of JP 2018-173662 A (Furi Gen).

Re claims 1-4, 7-13, and 19-20, Shimano et al. teaches Fig. 1 and associated text a

Shimano et al. teaches Fig. 1 and associated text a hard coat, substrate base, cured product of cycloolefin polymer [95] and [145-146] of meth acryloyloxy group (same as applicant's [90-95, 145-146] for ensuring a cured product and no cracking [217]. Inorganic particles are taught [15], Shimano (per claim 2).
See [210-222], Shimano – PET and cycloolefin structure (per claim 8). Re claims 19-20, see [210, 221-222]). Re claim 21, see [224] urethane acrylate.

All properties not explicit (i.e. peak area ratio...) are inherent given the same materials and structure used.

Further to product by process limitations (per claims 9 and 11) of before heat treatment and after curing, these are process limitations in a product claim and not germane to patentability.

Further re claim 1, Shimano doesn't teach amorphous inorganic fine particles on each surface as claimed.

Furi Gen teaches fumed silica on a side (see under Inorganic fine particles heading, pp. 9-11) to aid in excellent blackness and anti-glare properties.

It would have been obvious to one having ordinary skill in the art to have modified the inorganic skin film of Shiamo and incorporated Furi Gen's amorphous inorganic fine particles on each side of the base film for excellent blackness and anti-glare properties. The duplication of parts is generally recognized as being within the level of ordinary skill in the art, absent unexpected results. Providing additional clay-polymer composite barrier coatings would have achieved expected results such as providing more barrier properties. Mere duplication of parts has no patentable

significance unless a new and unexpected result is produced. MPEP 2144.04 (VI) B. In re Harza, 124 USPQ 378,380 (CCPA 1960).

Claims 5-6, 14-18, and 22-23 are rejected under 35 U.S.C. 103 as being unpatentable over US 20110217541 to Shimano et al. in view of JP 2018-173662 A (Furi Gen) and further in view of EP 2498130 A2 Shitabatake.

Shimano essentially teaches the claimed invention.

Re claims 5-6 and 14-18, Shimano doesn't teach the exact ranges but are overlapping. See [481], See Shimano – overlapping ranges (per claim 5). See [50-54], Shimano – overlapping ranges (per claims 6 and 14). Re claims 15-18, see [50, 57], Shimano overlapping ranges.

Note that overlapping ranges have this application: In the case where the claimed ranges "overlap or lie inside ranges disclosed by the prior art" a *prima facie* case of obviousness exists. *In reWertheim,* 541 F.2d 257, 191 USPQ 90 (CCPA 1976); *In reWoodruff,* 919 F.2d 1575, 16 USPQ2d 1934 (Fed. Cir. 1990). See MPEP 2144.05.

It would have been obvious to one of ordinary skill in the art at the effective filing date time the invention was made to have selected from the overlapping portion of the range taught by the reference effecting functionality because overlapping ranges have been held to establish prima facie obviousness. MPEP 2144.05.

Re claims 22-23, Shimano doesn't teach the materials as claimed but teaches silica, [224] urethane acrylate and [90] methacrylol.

Shitabatake discloses a similar curable composition [10] of excellent pattern with excellent heat/cold shock resistance having the [23] same (meth)acryloyl, amorphous

[235, 237-238] silica for good profile, and [257], [261] elastomer urethane and acrylic blend is within the teaching for improved heat resistance [256].

It would have been obvious to one of ordinary skill in the art at the effective filing date time the invention was made to have selected only the materials of Shimano in the similar cured composition of Shimano for at least excellent patterns and shock resistance.

References of Interest

The remaining references listed on form(s) 892 and/or 1449 have been reviewed by the examiner and are considered to be cumulative to or less material than the prior art references relied upon in the rejection above.

US 20180361719 to Kikuchi teaches a hard coat [153] where radically polymerizable groups include (meth)acryloyloxy and vinyl, of which (meth)acryloyloxy is generally employed.

Response to Arguments

Applicant's arguments filed 08/17/2026 have been fully considered but they are not persuasive. Note the spherical silica of Shimano doesn't teach away from the amorophous silicia teaching of the new reference. See action above.

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to TAMRA L. DICUS whose telephone number is (571)272-2022. The examiner can normally be reached M-F 8:00 am 4:00 pm.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Callie Shosho can be reached on 571-272-1123. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of published or unpublished applications may be obtained from Patent Center. Unpublished application information in Patent Center is available to registered users. To file and manage patent submissions in Patent Center, visit: https://patentcenter.uspto.gov. Visit https://www.uspto.gov/patents/apply/patent-center for more information about Patent Center and https://www.uspto.gov/patents/docx for information about filing in DOCX format. For additional questions, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

TAMRA L. DICUS
Primary Examiner
Art Unit 1787

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Primary Examiner, Art Unit 1787